

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech. (CL) Examination****March-April, 2018****CL203.01 Fluid Mechanics I****Date: 04.04.2018, Wednesday****Time: 01:30 p.m. To 4:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Define the following [07]**

uniform flow, steady flow, kinetic energy correction factor, law of conservation of mass, stream line, path line, laminar flow

Q - 2(a) Water is flowing in a fire hose with a velocity of 1.0 m/s and a pressure of 200000 Pa. [05]

At the nozzle the pressure decreases to atmospheric pressure (101300 Pa), there is no change in height. Use the Bernoulli equation to calculate the velocity of the water exiting the nozzle.

(b) Attempt any three questions [09]

- 1 An oil of 5m^3 weights 4000 kg (f). Calculate the specific weight and Specific gravity of this oil.
- 2 A 25 cm diameter pipe carried oil of specific gravity 0.8 at the rate 120 lit/sec. The pressure at point 'P' 3.5 m above datum line is 19.62 kN/m^2 (gauge) calculate the total energy at point 'P' in meters of oil.
- 3 Obtain the expression of continuity equation for a three dimensional incompressible flow.
- 4 A stream function in two dimensional flow is $\psi = 2xy$. Show that the flow is irrotational and determine the corresponding velocity potential ϕ .
- 5 A three dimensional flow field described by $V = (x^2+z^2) i + (x^2+z^2) j + (x^2+y^2) k$.
Compute the components of velocity and total velocity at point (1, 2, 3)

Q - 3(a) A 45° bend is connected in a pipe line, the diameter at the inlet and outlet of the bend is [05]
600 and 300 mm respectively. If the rate of flow is 600 lit/s and pressure intensity at inlet of bend is 8.829 N/cm^2 , find the force exerted by water to the bend.

(b) Attempt any three questions [09]

- 1 A jet of water is issued from a nozzle at velocity 13 m/sec from a point 5 m from a vertical wall of 7.5 m height. Find the angle of projection of the jet to just clear a wall.

- 2 Enlist the forces influencing the motion of a fluid flow. Which forces are considered in the expression of Euler's equation.
- 3 Derive Bernoulli's equation for flow of an incompressible frictionless fluid.
- 4 Water is flowing through a pipe having diameter 300 mm and 200 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 24.525 N/cm^2 and the pressure at the upper end is 9.81 N/cm^2 . Determine the difference in datum head if the rate of flow through pipe is 40 lit/s.
- 5 Distinguish between rotational and irrotational flow.

SECTION – II

Q - 4 Match the sets A and B

A	B	[07]
1. Metacentric height	(a) weight of displaced volume	
2. Centre of Gravity below Centre of Buoyancy	(b) CG of displaced volume	
3. Centre of buoyancy	(c) Stability	
4. Buoyant force	(d) Always stable	
5. The center of gravity is	(e) Hydrostatics	
6. Study of pressure exerted by water at rest	(f) Equal	
7. In 'Pipes in Series' discharge is	(g) Centre of weight of body	

- Q – 5(a)** (i) What is manometer? How they are classified? Briefly explain measurement of pressure with any one type of manometer with the help of neat sketch. [04]
- (ii) A simple manometer is used to measure the pressure of oil of specific gravity 0.8 flowing in a pipe line. Its right limb is open to the atmosphere and the left limb is connected to the pipe. The center of pipe is 9 cm below the level of mercury (sp. gravity 13.6) in the right limb. If the difference of mercury level in the two limb is 15 cm, determine the absolute pressure of the oil in the pipe in kg/cm^2 . [03]

OR

- (a)** (i) State and explain Pascal's law. [04]
- (ii) Determine total pressure and center of pressure on an isosceles triangular plate of base 4m and height 4m when it is immersed vertically into oil of sp. Gr. 0.9. the base of plate coincide with free surface of oil [03]
- (b)** (i) What is Buoyancy? Differentiate between center of buoyancy and center of gravity. [04]

(ii) With the neat sketch explain the condition of equilibrium of a floating body.

- Q - 6 (a)** (i) Give practical examples of laminar flow and explain its characteristics. [03]
 (ii) Explain the occurrence of Water Hammer in brief [04]

OR

- (a)** (i) Explain briefly following : [04]
 (1) Hydraulic Gradient Line and Total Energy Line
 (2) Pipes in Series and Pipes in Parallel.
- (ii) A horizontal pipe 125 mm diameter is joined by sudden enlargement to a 200 mm diameter pipe. Water is flowing through it at the rate $0.075 \text{ m}^3/\text{s}$. Find pressure difference in two pipes in m of water. [03]
- (b)** (i) Derive formula for finding discharge over a rectangular-Notch. [03]
 (ii) During an experiment in a laboratory a rectangular notch is used to measure the flow rate of water in an open channel. The width of the rectangular notch is 3 m and the head of water causing the flow is 0.5 m. Calculate the discharge assuming the coefficient of discharge of the notch as 0.6. [02]
 (iii) A venturimeter is used to measure liquid flow rate of 7500 litres per minute. The difference in pressure across the venturimeter is equivalent to 8 m of the flowing liquid. The pipe diameter is 19 cm. Calculate the throat diameter of the venturimeter. Assume the coefficient of discharge for the venturimeter as 0.96. [02]
